1 Discrete Time Signals and System

Introduction

Signal

A signal is any time varying physical phenomenon, which intends to convey some information, for example, traffic signals, human voice, telephone or television signals, sensex movement on a stock exchange. etc.

A signal is formally defined as a function of one or more variables that conveys information on the nature of a physical phenomenon.

System

A system is an entity, which manipulates one or more input signals to generate new or desired output signals.

Discrete Time Signal

A 'Discrete Time (DT) signal' is defined or specified at discrete values or instants of time (t) and the value is not defined for remaining time. A DT signal is represented as x[n] or y[n] where 'n' is discrete time. Note that 'n' takes only integral values. The daily closing prices of Facebook shares, etc., is an example of discrete time signal [Fig. 1.1 (a)–(c)].

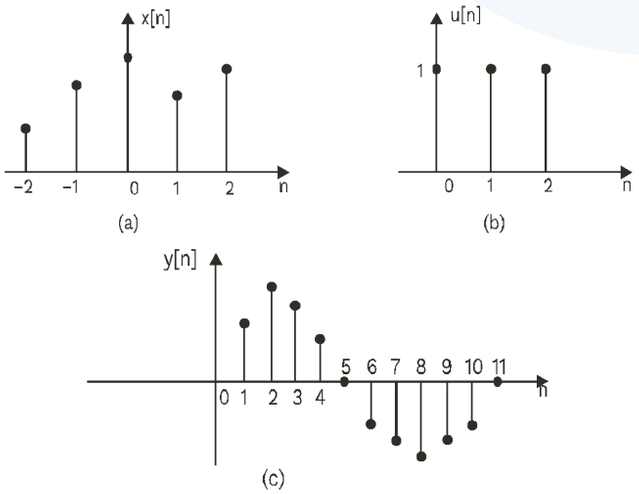


Fig. 1.1 Discrete Time Signals

Representation of Discrete Time Signals

1. Function representation

$$x(n) = \begin{cases} 1 & \text{for } n = 1, 3 \\ 4 & \text{for } n = 2 \\ 0 & \text{else where} \end{cases}$$

2. Tabular representation

n	–2	–1	0	1	2
x(n)	0	1	0	2	3

3. Sequence representation

$$x(n) = \{1, 2, 0, 1\}$$
$$\uparrow$$
$$n = 0$$

Classification of Discrete Time Signals

1. Even and odd signal
2. Periodic and aperiodic signal
3. Energy and power signal

Even and Odd Signals

A signal x[n] is even, if

$x[-n] = x[n]$

i.e., a time reversed signal is identical to the original signal. In other words, even signals are symmetric about vertical (i.e., amplitude) axis or time origin.

Example:

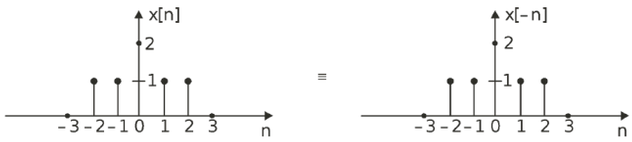


Fig. 1.2 Even Signals

A signal x[n] is odd, if

$x[-n] = -x[n]$

Note: n = 0 $x[-0] = -x[0]$ $x[0] = 0$

i.e., odd signals must be necessarily zero at time origin.

Odd signals are anti-symmetric about the time origin.

Example:

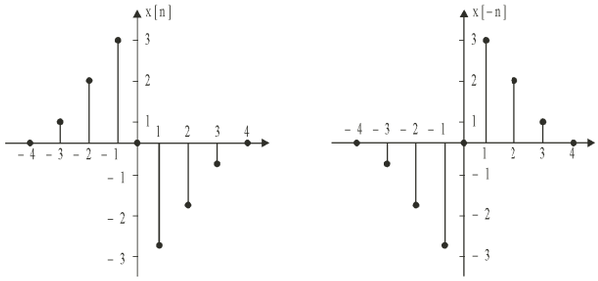


Fig. 1.3 Odd Signals

Even and odd parts of x[n]

$$x_e[n] = \frac{1}{2}[x[n] + x[-n]]$$

& $$x_0[n] = \frac{1}{2}[x[n] - x[-n]]$$

Properties of Even and Odd Signal

1. Even signal × Even signal = Even signal
   Odd signal × Odd signal = Even signal
   Even signal × Odd signal = Odd signal × Even signal = Odd signal
2. If x(n) is an even signal, then

$$\sum_{n=-\infty}^{\infty} x(n) = x(0) + 2\sum_{n=1}^{\infty} x(n)$$

Proof: Consider x(n) is an even signal, then

$$\sum_{n=-\infty}^{\infty} x(n) = \sum_{n=-\infty}^{-1} x(n) + x(0) + \sum_{n=1}^{\infty} x(n)$$

$$= \sum_{n=1}^{\infty} x(-n) + x(0) + \sum_{n=1}^{\infty} x(n)$$

$$= \sum_{n=1}^{\infty} x(n) + x(0) + \sum_{n=1}^{\infty} x(n)$$

[$\because$ x(n) is an even signal]

$$\sum_{n=-\infty}^{\infty} x(n) = x(0) + 2\sum_{n=1}^{\infty} x(n)$$

3. If x(n) is an odd signal, then

$$\sum_{n=-\infty}^{\infty} x(n) = 0$$

Proof: Consider x(n) is an odd signal, then

$$\sum_{n=-\infty}^{\infty} x(n) = \sum_{n=-\infty}^{-1} x(n) + \sum_{n=0}^{\infty} x(n)$$

$$= \sum_{n=1}^{\infty} x(-n) + x(0) + \sum_{n=1}^{\infty} x(n)$$

$$= -\sum_{n=1}^{\infty} x(n) + 0 + \sum_{n=1}^{\infty} x(n)$$

[$\because$ x(n) is odd signal $\therefore$ x(−n) = −x(n), x(0) = 0]

$$\sum_{n=-\infty}^{\infty} x(n) = 0$$

4. If x(n) is an arbitrary signal and its even and odd parts are represented by $x_e(n)$ and $x_0(n)$ respectively, then

$$\sum_{n=-\infty}^{\infty} x^2(n) = \sum_{n=-\infty}^{\infty} x_0^2(n) + \sum_{n=-\infty}^{\infty} x_0^2(n)$$

Proof: If x(n) is an arbitrary signal, x(n) = $x_e(n) + x_0(n)$

$$\sum_{n=-\infty}^{\infty} x^2(n) = \sum_{n=-\infty}^{\infty} [x_e(n) + x_0(n)]^2$$

$$= \sum_{n=-\infty}^{\infty} [x_e^2(n) + x_0^2(n) + 2x_0(n)x_0(n)]$$

$$= \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_0^2(n) + 2\sum_{n=-\infty}^{\infty} x_e(n)x_0(n)$$

$$= \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_0^2(n)$$

$$+ 2\left[\sum_{n=-\infty}^{-1} x_e(n)x_0(n) + \sum_{n=0}^{\infty} x_e(n)x_0(n)\right]$$

$$= \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_0^2(n)$$

$$+ 2\left[\sum_{n=1}^{\infty} x_e(-n)x_0(-n) + \sum_{n=0}^{\infty} x_e(n)x_0(n)\right]$$

$$= \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_0^2(n)$$

$$+ 2\left[\sum_{n=1}^{\infty} x_e(-n)x_0(-n) + x_e(0)x_0(0) + \sum_{n=1}^{\infty} x_e(n)x_0(n)\right]$$

$x_e(n) = x_e(-n), x_0(n) = -x_0(-n)$ and $x_0(0) = 0$,

then we get

$$= \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_0^2(n) + 2\left[-\sum_{n=1}^{\infty} x_e(n)x_0(n) + (0) + \sum_{n=1}^{\infty} x_e(n)x_0(n)\right]$$

$$= \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_0^2(n) + 2(0)$$

$$\Rightarrow \quad \sum_{n=-\infty}^{\infty} x^2(n) = \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_0^2(n)$$

For Complex Valued Signal

- If x(n) is a complex valued signal, then for even conjugate or conjugate symmetric signal

  $x(n) = x \times (-n)$

- For odd conjugate or conjugate anti-symmetric signal

  $x(n) = -x \times (-n)$

Decomposition of any arbitrary signal x(n), into even and odd conjugate components $x_{ec}(n)$ and $x_{oc}(n)$ respectively:

$$x_{ec}(n) = \frac{x(n) + x \times (-n)}{2}$$

and

$$x_{oc}(n) = \frac{x(n) - x \times (-n)}{2}$$

Solved Examples

1. Sketch the even and odd components of the signal x[n] shown in the figure below.

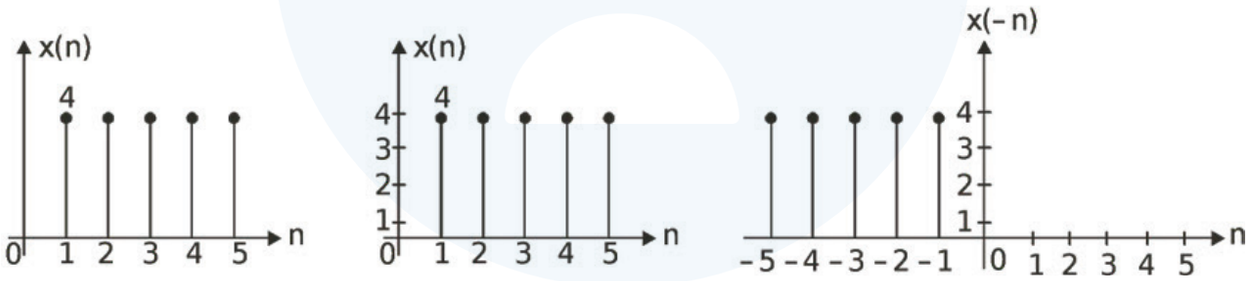


Fig. 1.4

Solution:

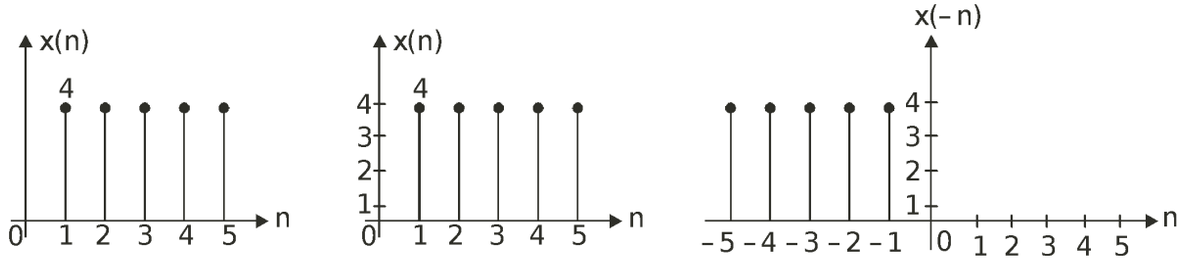


Fig. 1.5

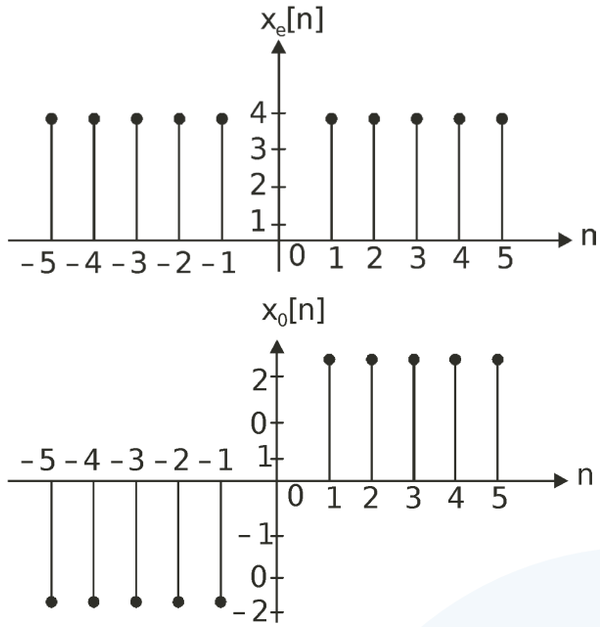


Fig. 1.6

Periodic and Non-periodic (or Aperiodic) Signals

A discrete time signal is periodic with N, where N is a positive integer

if $x[n + N] = x[n]$ for all n

The smallest value of N is called fundamental period N_0 of x[n].

Note:

1. A discrete time signal obtained by uniform sampling of a periodic continuous time signal may not be periodic.
2. The sum of two periodic sequence (discrete time) is always periodic.

Solved Examples

2. Find the condition for periodicity of complex exponential discrete time (DT) signal

$x[n] = e^{j\Omega_0 n}$

Solution:

$x[n + N] = x[n]$

$\Rightarrow \quad e^{j\Omega_0(n+N)} = e^{j\Omega_0 n}$

$\Rightarrow \quad e^{j\Omega_0 N} = 1$

$\Rightarrow \quad \Omega_0 N = 2k\pi$

k = Positive integer ($\because \Omega_0$, N are positive)

$\Rightarrow \quad \frac{\Omega_0}{2\pi} = \frac{k}{N}$ = Rational number

Thus, x[n] is periodic, if $\frac{\Omega_0}{2\pi}$ is a rational number.

Note: N is not necessarily the fundamental period of x[n]

3. Determine whether the following discrete time signals are periodic or not, and if periodic determines the fundamental period.

a. $x[n] = \sin(5\pi n)$

b. $x[n] = \cos(4n)$

c. $x[n] = \sin\left(\frac{2\pi n}{3}\right) + \cos\left(\frac{2\pi n}{5}\right)$

d. $x[n] = \cos\left(\frac{\pi}{2}n\right)\cos\left(\frac{\pi}{4}n\right)$

Solution:

a. $x[n] = \sin(5\pi n)$

$\omega_0 = 5\pi$

$\frac{\omega_0}{2\pi} = \frac{5\pi}{2\pi} = 2.5$ = Rational number

$N = \left(\frac{2\pi}{\omega}\right)m = \left(\frac{2\pi}{5\pi}\right)m$

$= \left(\frac{2}{5}\right)m$

$= 2 \ (m = 5)$

$x[n] = \sin(5\pi n)$ is periodic with fundamental period N = 2

b. $x[n] = \cos(4n)$

$\omega_0 = 4$

$\frac{\omega_0}{2\pi} = \frac{4}{2\pi}$ = Irrational number

$x[n] = \cos(4n)$ is aperiodic signal

c. $x[n] = \sin\left(\frac{2\pi n}{3}\right) + \cos\left(\frac{2\pi n}{5}\right)$

$\omega_1 = \frac{2\pi}{3}$

$\omega_2 = \frac{2\pi}{5}$

$\frac{\omega_1}{2\pi} = \frac{2\pi/3}{3} = \frac{1}{3}$ = Rational number

$\frac{\omega_2}{2\pi} = \frac{2\pi/5}{2\pi} = \frac{1}{5}$ = Rational number

$$N_1 = \left(\frac{2\pi}{\omega_1}\right)m = \left(\frac{2\pi}{2\pi}\cdot 3\right)m = 3m = 3(m=1)$$

$$N_2 = \left(\frac{2\pi}{\omega_2}\right)K = \left(\frac{2\pi}{2\pi}\cdot 5\right)K = 5K = 5(K=1)$$

$$\frac{N_1}{N_2} = \frac{3}{5} = \text{Rational number}$$

x[n] is periodic

Fundamental time period x[n] is N = LCM of (3 and 5)

N = 15

d. $x[n] = \cos\left(\frac{\pi}{2}n\right)\cos\left(\frac{\pi}{4}n\right)$

$$= \frac{1}{2}\cos\left(\frac{3\pi}{4}n\right) + \frac{1}{2}\cos\left(\frac{\pi}{4}n\right)$$

$$\omega_1 = \frac{3\pi}{4}$$

$$\omega_2 = \frac{\pi}{4}$$

$$N_1 = \left(\frac{2\pi}{\omega_1}\right)m = \left(\frac{2\pi}{3\pi}4\right)m = \frac{8}{3}m = 8(m=3)$$

$$N_2 = \left(\frac{2\pi}{\omega_2}\right)K = \left(\frac{2\pi}{\pi}4\right)K = 8K = 8(K=1)$$

$$\frac{N_1}{N_2} = \frac{8}{8} = 1 = \text{Rational number}$$

x[n] is periodic

N = LCM of (8 and 8)

N = 8

4. Let $x_1[n]$ and $x_2[n]$ be period sequence with fundamental periods N_1 and N_2, respectively. Under what conditions is the sum $x[n] = x_1[n] + x_2[n]$ periodic, and what is the fundamental period of x[n] if it is periodic?

Solution:

$x_1[n]$ and $x_2[n]$ are periodic with fundamental periods N_1 and N_2, respectively. We have:

$$x_1[n] = x_1[n+N_1] = x_1[n+mN_1]$$

$$x_2[n] = x_2[n+N_2] = x_2[n+kN_2]$$

Where, m and k are positive integers.

$$x[n] = x_1[n] + x_2[n]$$

$$x[n+N] = x_1[n] + x_2[n]$$

$$x[n+N] = x_1[n+mN_1] + x_2[n+kN_2]$$

Thus, we must have:

$$mN_1 = kN_2 = N$$

$$\frac{N_1}{N_2} = \frac{k}{m} = \text{Rational number}$$

Therefore, the sum of two sequence, i.e., $x[n] = x_1(n) + x_2(n)$ is periodic, if

$$\frac{N_1}{N_2} = \text{Rational number}$$

Where, $N_2 \rightarrow$ Fundamental time period of $x_1(n)$

$N_2 \rightarrow$ Fundamental time period of $x_2(n)$

And fundamental time period of x(n) is N = LCM of $(N_1 \& N_2)$

Energy and Power Signals

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

and

$$P_x = \lim_{N\to\infty} \frac{1}{(2N+1)} \sum_{n=-N}^{N} |x[n]|^2$$

Energy signal: A signal with finite energy is called an energy signal.

Mathematically: $0 < E < \infty$, $P = 0$

Power signal: A signal with finite and non-zero power is a power signal.

Mathematically: $0 < P < \infty$, $E = \infty$

Note:

1. A signal with finite energy has zero power. Hence, an energy signal cannot be a power signal.
2. A signal with finite power has infinite energy. Hence, a power signal cannot be an energy signal.
3. A signal may be neither an energy signal nor a power signal, for example, ramp signal.
4. All practical signals have finite energy and are, therefore, energy signals. A true power signal is impossible as it must have infinite duration and infinite energy.

Solved Examples

5. Determine whether the following signals are energy signals or power signals, or neither energy nor power signals.

a. $x[n] = \cos\left[\frac{\pi}{4}n\right]$

b. $x[n] = u[n] - u[n-6]$

c. $x[n] = n\,u[n]$

d. $x[n] = r[n] - r[n-4]$

e. $x[n] = (-0.5)^n\,u[n]$

f. $x(n) = 2e^{j3n}$

Solution:

a. $x[n] = \cos\left[\frac{\pi}{4}n\right]$

$$|x[n]|^2 = \cos^2\left[\frac{\pi}{4}n\right] = \frac{1+\cos\left[\frac{\pi}{2}n\right]}{2}$$

$$E_x = \sum_{n=-\infty}^{\infty}|x[n]|^2 = \sum_{n=-\infty}^{\infty}\frac{1+\cos\left[\frac{\pi}{2}n\right]}{2}$$

$$= \sum_{n=-\infty}^{\infty}\frac{1}{2} + \sum_{n=-\infty}^{\infty}\frac{1}{2}\cos\left[\frac{\pi}{2}n\right]$$

$E_x = \infty$

$$P_x = \underset{N\to\infty}{Lt}\frac{1}{2N+1}\sum_{n=-N}^{N}|x[n]|^2$$

$$= \underset{N\to\infty}{Lt}\frac{1}{2N+1}\sum_{n=-N}^{N}\frac{1+\cos\left[\frac{\pi}{2}n\right]}{2}$$

$$= \underset{N\to\infty}{Lt}\frac{1}{2N+1}\sum_{n=-N}^{N}\frac{1}{2} + \frac{1}{2N+1}\sum_{n=-N}^{N}\frac{1}{2}\cos\left[\frac{\pi}{2}n\right]$$

$$= \underset{N\to\infty}{Lt}\left(\frac{1}{2N+1}\right)\frac{(2N+1)}{2} + 0$$

$P_x = \frac{1}{2}$

$x[n] = \cos\left[\frac{\pi}{4}n\right]$ is power signal with $P_x = \frac{1}{2}$

b.

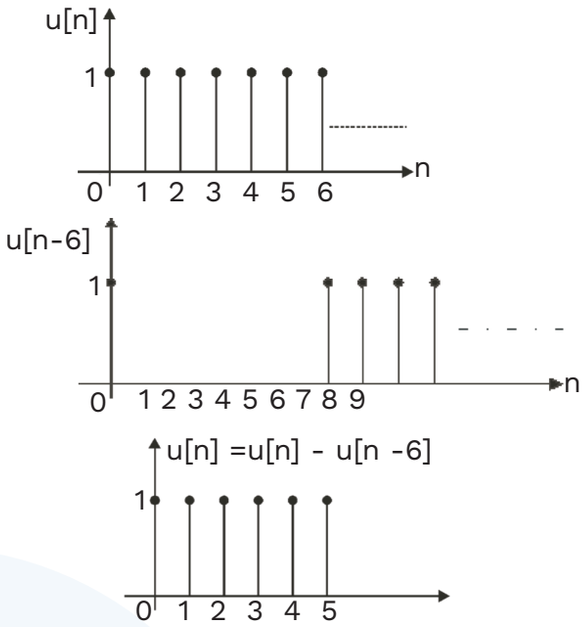


Fig. 1.7

$$x[n] = \begin{cases}1, & 0 \le n \le 5\\ 0, & \text{Otherwise}\end{cases}$$

$$E_x = \sum_{n=-\infty}^{\infty}|x[n]|^2$$

$$= \sum_{n=0}^{5}1 = 6$$

$E_x = 6$

$$P_x = \underset{N\to\infty}{Lt}\frac{1}{2N+1}\sum_{n=-N}^{N}|x[n]|^2$$

$$= \underset{N\to\infty}{Lt}\frac{1}{2N+1}\sum_{n=0}^{5}1$$

$$= \underset{N\to\infty}{Lt}\frac{1}{2N+1}6 = 0$$

$P_x = 0$

$x[n] = u[n] - u[n-6]$ is energy signal with $E_x = 6$

c. $x[n] = nu[n]$

$$= \begin{cases}n & n \ge 0\\ 0 & n < 0\end{cases}$$

$$E_x = \sum_{n=-\infty}^{\infty}|x[n]|^2 = \sum_{n=0}^{\infty}n^2$$

$E_x = \infty$

$$P_x = \underset{N\to\infty}{Lt}\frac{1}{2N+1}\sum_{n=-N}^{N}|x(n)|^2$$

$$= \underset{N\to\infty}{Lt}\frac{1}{2N+1}\sum_{n=0}^{N} n^2$$

$$= \underset{N\to\infty}{Lt}\left(\frac{1}{2N+1}\right)\frac{N(N+1)(2N+1)}{6}$$

$P_x = \infty$

x[n] = nu[n] is neither energy nor power signal

d. x[n] = r[n] – r[n – 4]

$$= \begin{cases} n & 0 \le n \le 4 \\ 4 & n \ge 5 \end{cases}$$

$$E_x = \sum_{n-\infty}^{\infty} |x[n]|^2$$

$$= \sum_{n=0}^{4} n^2 + \sum_{n=5}^{\infty} (4)^2$$

$$= 30 + \infty$$

$E_x = \infty$

$$P_x = \underset{N\to\infty}{Lt}\frac{1}{2N+1}\sum_{n=-N}^{N} |x[n]|^2$$

$$= \underset{N\to\infty}{Lt}\frac{1}{(2N+1)}\left[\sum_{n=0}^{4} n^2 + \sum_{n=5}^{N} (4)^2\right]$$

$$= \underset{N\to\infty}{Lt}\frac{1}{(2N+1)}\left[30 + 16(N-4)\right]$$

$$= \underset{N\to\infty}{Lt}\frac{30}{(2N+1)} + \underset{N\to\infty}{Lt}\frac{16(N-4)}{(2N+1)}$$

$$= 0 + \underset{N\to\infty}{Lt}\frac{16N\left(1-\frac{4}{N}\right)}{2N\left(1+\frac{1}{2N}\right)}$$

$P_x = 8$

x[n] = r[n] – r[n – 4] is power signal with P_x = 8

e. $x[n] = (-0.5)^n u[n]$

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$= \sum_{n=0}^{\infty} (0.25)^n = \frac{1}{1-0.25} = \frac{1}{1-\frac{1}{4}}$$

$$E_x = \frac{4}{3}$$

$$P_x = \lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N} |x(n)|^2$$

$$= \lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=0}^{N} (0.25)^2$$

$$= \lim_{N\to\infty}\frac{1}{2N+1}\left[\frac{1-(0.25)^{N+1}}{1-0.25}\right]$$

$P_x = 0$

Therefore, $x[n] = (-0.5)^n$ u(n) is energy signal with $E_x = \frac{4}{3}$

f. $x[n] = 2e^{j3n}$

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$= \sum_{n=-\infty}^{\infty} \left|2e^{j3n}\right|^2 = \sum_{n=-\infty}^{\infty} 4 = \infty$$

$E_x = \infty$

$$P_x = \lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N} |x(n)|^2$$

$$= \lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N} \left|2e^{j3n}\right|^2$$

$$P_x = \lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N} 4$$

$$P_x = \lim_{N\to\infty}\frac{4}{(2N+1)}\sum_{n=-N}^{N} 1$$

$$= \lim_{N\to\infty}\left(\frac{4}{2N+1}\right)(2N+1)$$

$P_x = 4$

Therefore $x[n] = 2e^{j3n}$ is power signal with $P_x = 4$

6. Find whether the signal

$$x[n] = \begin{cases} n^2 & 0 \le n \le 3 \\ 10-n & 4 \le n \le 6 \\ n & 7 \le n \le 9 \\ 0 & \text{otherwise} \end{cases}$$

is a power signal or an energy signal. Also, find the energy and power of the signal.

Solution:

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

$$E_x = \sum_{n=0}^{3} n^4 + \sum_{n=4}^{6} (10-n)^2 + \sum_{n=7}^{9} n^2$$

$E_x = 369$

$$P_x = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2$$

$$= \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=0}^{3} n^4 + \sum_{n=4}^{6} (10-n)^2 + \sum_{n=7}^{9} n^2$$

$P_x = 0$

x[n] is energy signal with $E_x = 369$

7. Identify whether each of the following signals are energy or power signals.

a. $x[n] = \begin{cases} \sin(\pi n) & -7 \le n \le 7 \\ 0 & \text{otherwise} \end{cases}$

b. $x[n] = \begin{cases} \cos(\pi n) & -6 \le n \le 6 \\ 0 & \text{otherwise} \end{cases}$

c. $x[n] = \begin{cases} \cos(\pi n) & n \ge 0 \\ 0 & \text{otherwise} \end{cases}$

Solution:

a. Note that

x[n] = 0, for all n

$\because$ $\sin n\pi = 0$, for all integer n

Hence, x[n] is a zero signal.

b. $\because$ x[n] is a finite duration signal. Hence, it cannot be a power signal.

Normalised energy content in x[n] is:

$$E = \sum_{n=-6}^{6} \cos^2(n\pi)$$

$$= 2\sum_{n=1}^{6} \cos^2(\pi n) + \cos^2(\pi)$$

$$= 2\{(-1)^2 + (1)^2 + (-1)^2 + (1)^2 + (-1)^2 + (1)^2\} + 1$$

E = 13 = A finite value (non-zero)

$\therefore$ x[n] is an energy signal.

c. $x[n] = \begin{cases} \cos(\pi n)\,; & n \ge 0 \\ 0\,, & \text{otherwise} \end{cases}$

$\because$ $\cos n\pi = (-1)^n$

$$\lim_{N\to\infty} \frac{1}{(2N+1)} \sum_{n=-N}^{N} (x[n])^2$$

$$\therefore P = = \lim_{N\to\infty} \frac{1}{(2N+1)} \sum_{n=0}^{N} [(-1)^n]^2$$

$$P = \lim_{N\to\infty} \frac{1}{(2N+1)} (N+1)$$

$$P = \lim_{N\to\infty} \frac{1+\frac{1}{N}}{2+\frac{1}{N}}$$

$P = \frac{1}{2}$ = Finite non-zero value

Hence, x[n] is a power signal.

Effects of Operations Performed on Independent Variable (Time) and Dependent Variable (Amplitude)

Time Reversal or Reflection

y[n] = x[−n]

y[n] is obtained by taking reflection of x[n] about amplitude axis.

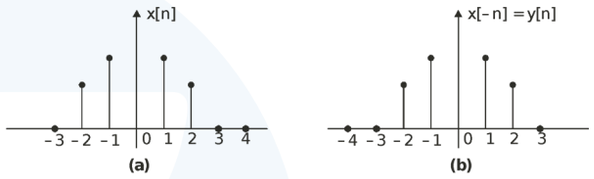


Fig. 1.8

Time Scaling

y[n] = x[an]

Where, 'a' is known as scaling factor.

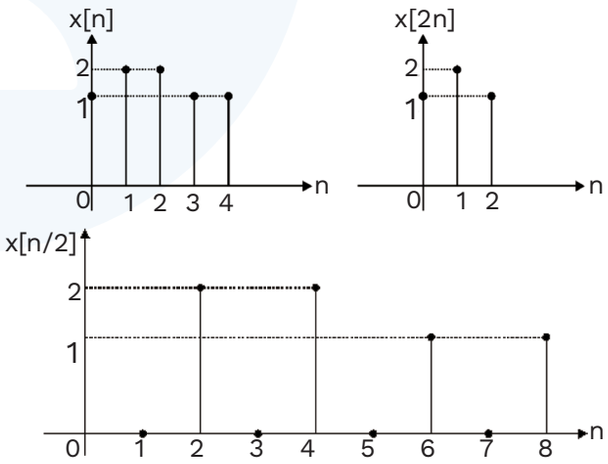


Fig. 1.9

Note: For a < 0, use time reversal operation after scaling.

Time Shifting

A signal x(t) may be shifted in time by replacing the independent variable n by $n - n_0$, where n_0 is known as the shifting factor.

Time shifted or delayed version of x[n] is given by

$y[n] = x[n - n_0]$

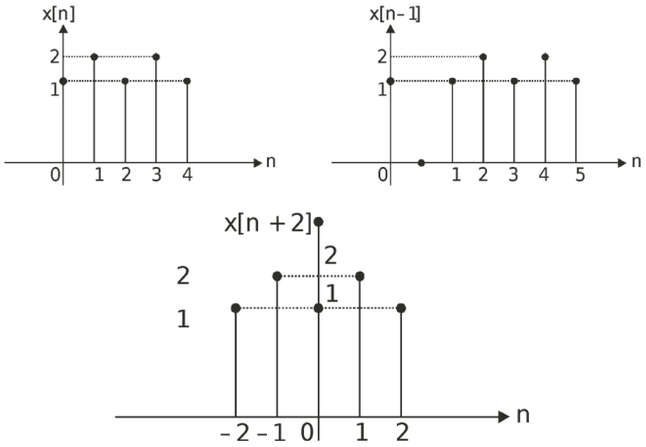


Fig. 1.10

Operation on Dependent Variables

For Discrete Time Signal

1. Addition $y[n] = x_1[n] + x_2[n]$
2. Multiplication $y[n] = x_1[n]\, x_2[n]$
3. Differencing

   First forward difference $y[n] = x[n + 1] - x[n]$

   First backward difference $y[n] = x[n] - x[n - 1]$
4. Accumulation $y[n] = \sum_{K=-\infty}^{n} x[K]$

Elementary or Basic Signals

- Unit step function or unit amplitude step function
- Unit impulse function
- Signum function
- Ramp function
- Rectangular function
- Exponential and sinusoidal functions
- Sinc function
- Dirichlet function

Note: Function and signal are synonymous to each other.

Unit Step Function or Unit Amplitude Step Function

- The discrete time unit step sequence (signal) u[n] is defined as:

$$u[n] = \begin{cases} 1 & , \ n \geq 0 \\ 0 & , \ n < 0 \end{cases}$$

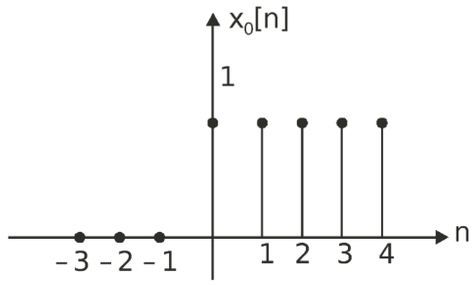


Fig. 1.11

- Shifted or delayed unit step signal is:

$$u[n - k] = \begin{cases} 1 & , \ n \geq 0 \\ 0 & , \ n < 0 \end{cases}$$

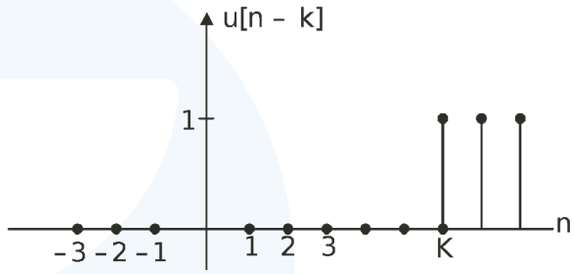


Fig. 1.12

Unit Impulse Sequence

- $\delta[n]$ is discrete time counter part of $\delta(t)$. $\delta[n]$ is also known as 'unit sample' sequence kronecker delta function.
- $\delta[n]$ is defined as:

$$\delta[n] = \begin{cases} 1 & , \ n = 0 \\ 0 & , \ n \neq 0 \end{cases}$$

- Shifted or delayed unit impulse (or sample) sequence is defined as:

$$\delta[n - k] = \begin{cases} 1 & , \ n = k \\ 0 & , \ n \neq k \end{cases}$$

Stem plots of $\delta[n]$ and $\delta[n - k]$

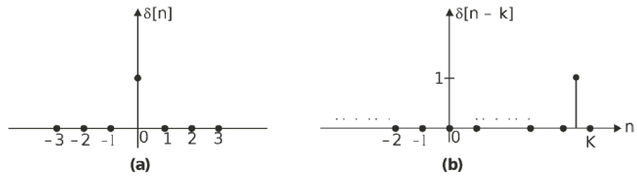


Fig. 1.13

Properties of δ[n]

- $\delta[an] = \delta[n]$ for any finite non-zero integer 'a'. Note that $\delta[n]$ does not follow scaling property of $\delta(t)$.
- Sampling property

$$\sum_{n=-\infty}^{\infty} \delta[n - n_0]x[n] = x[n_0]$$

Consequences

- $x[n]\delta[n] = x[0]\delta[n]$
- $x[n]\delta[n-k] = x[k]\delta[n-k]$

Relationship between δ[n] and u[n]

$$\delta[n] = u[n] - u[n-1]$$

$$\& \; u[n] = \sum_{k=-\infty}^{n} \delta[k]$$

Expression of Any Signal x[n] in Terms of δ[n]

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[k-n]$$

$$= \sum_{k=-\infty}^{\infty} x[k]\delta[-(n-k)]$$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

Signum Function

$$\text{sgn}[n] = \begin{cases} 1, & n > 0 \\ 0, & n = 0 \\ -1, & n < 0 \end{cases}$$

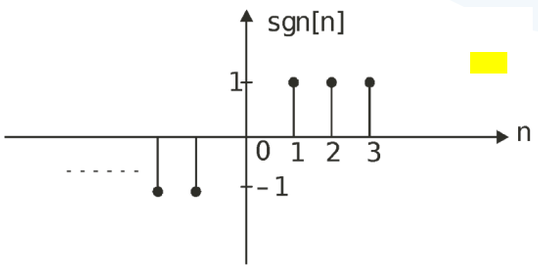


Fig. 1.14

Ramp Function

$$\text{ramp}[n] = r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

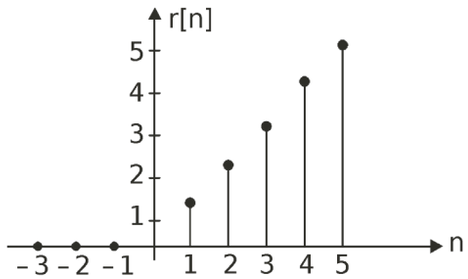


Fig. 1.15

Rectangular Function

$$\text{rect}_N[n] = \begin{cases} 1, & |n| \leq N \\ 0, & |n| > N \end{cases}$$

$$= u[n+N] - u[n-N-1]$$

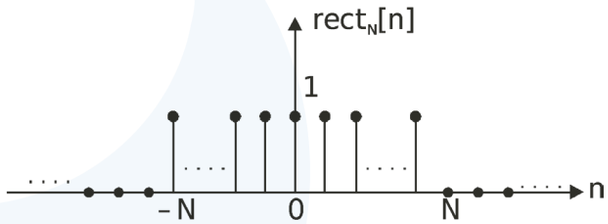


Fig. 1.16

Exponential and Sinusoidal Functions

Real Exponential sequences

$$x[n] = Aa^n, -\infty < n < \infty$$

where A, a are real parameters.

Case I: a > 1

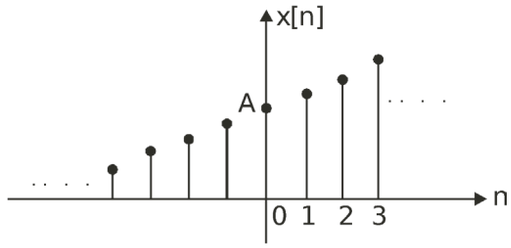


Fig. 1.17

Case II: 0 < a < 1

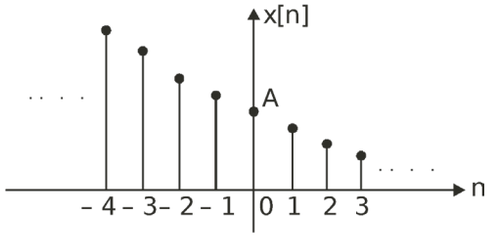


Fig. 1.18

Case III: a < –1

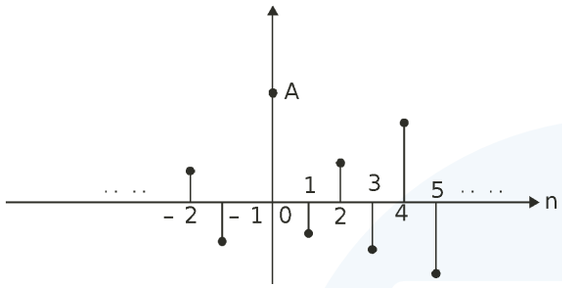


Fig. 1.19

Case IV: –1 < a < 0

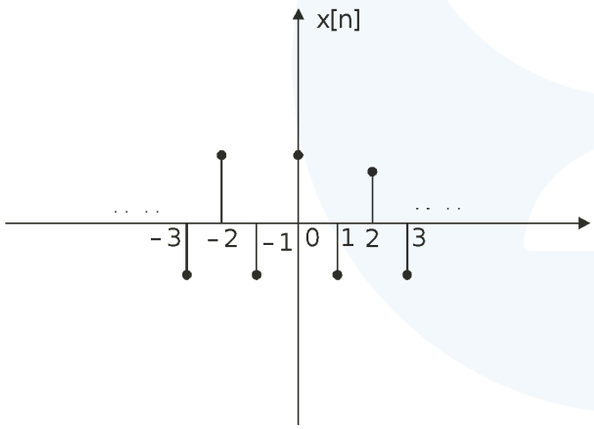


Fig. 1.20

Complex Exponential Sequence

$x[n] = e^{j\Omega_0 n}$

$x[n] = \cos\Omega_0 n + j\sin\Omega_0 n$

Condition for periodicity of $x[n] = e^{j\Omega_0 n}$

$\frac{\Omega_0}{2\pi} = \frac{k}{N}, k$ = Positive integer

= Rational number

Fundamental period (N_0) is given by

$N_0 = k\left(\frac{2\pi}{\Omega_0}\right)$ = a least positive integer

Note:

- $e^{j\omega_0 t}$ is periodic for all ω_0, while $e^{j\Omega_0 n}$ is not periodic for all Ω_0.
- $e^{j\omega_0 t}$ is distinct for distinct value of ω_0, while $e^{j\Omega_0 n}$ is not.

  $\because\ e^{j(\Omega_0+2\pi m)n} = e^{j\Omega_0 n}e^{j2\pi mn}$

  $= e^{j\Omega_0 n}\cdot 1 = e^{j\Omega_0 n}\cdot 1$

  $= e^{j\Omega_0 n}$

- Complex exponential sequence at Ω_0 is the same as that at frequencies. $\Omega_0 \pm 2\pi, \Omega_0 \pm 4\pi, \ldots\ldots$ Thus, generally Ω_0 is chosen as $0 \le \Omega_0 < 2\pi$ or $-\pi \le \Omega_0 < \pi$.

  (An important concept to be used extensively in discrete time signals and systems analysis).

Sinusoidal Sequence

- These are of the form

  $x[n] = A\cos(\Omega_0 n + \phi)$ or $B\sin(\Omega_0 n + \theta)$

Note: If n is dimensionless, then both Ω_0 & θ(or ϕ) have units of radians.

$x[n] = A\cos(\Omega_0 n + \phi)$

$= A\,\mathrm{Re}\left\{e^{j(\Omega_0 n+\phi)}\right\}$

All conclusions drawn for complex exponential sequence are valid for sinusoidal sequences as well.

Solved Examples

8. Identify if the following signals are periodic or aperiodic. If periodic, then find their fundamental period.

a. $x[n] = 14\sin\left(\frac{3\pi}{29}n + \frac{\pi}{3}\right)$

b. $x[n] = 7\sin(5n)$

c. $x[n] = 2\cos(5\pi n)$

Solution:

a. $\Omega_0 = \frac{3\pi}{29}$

$\frac{\Omega_0}{2\pi} = \frac{3\pi}{29\times 2\pi} = \frac{3}{58}$ = A rational number

$\therefore$ x[n] is periodic and its fundamental period

$N_0 = k\left(\frac{2\pi}{\Omega_0}\right)$

$= k\left(\frac{58}{3}\right)$

N_0 = 58; k = 3 (for least value)

b. x[n] = 7 sin(5n)

$\Omega_0 = 5$

$\therefore \frac{\Omega_0}{2\pi} = \frac{5}{2\pi}$ = Irrational number (due to π)

$\therefore$ x[n] is not periodic

c. x[n] = $2\cos(5\pi n)$

$\Omega_0 = 5\pi$

$\frac{\Omega_0}{2\pi} = \frac{5\pi}{2\pi} = \frac{5}{2}$ = Rational number

$\therefore$ x[n] is periodic

Fundamental period:

$N_0 = k\left(\frac{2\pi}{\Omega_0}\right)$

$= k\left(\frac{2}{5}\right)$

N_0 = 2; for k = 5

9. Consider the discrete time signal.

$$x[n] = 1 - \sum_{k=3}^{\infty} \delta[n-1-k]$$

Determine the values of the integers M and n_0 so that x[n] may be expressed as:

$x[n] = u[Mn - n_0]$

Solution:

$$x[n] = 1 - \sum_{k=3}^{\infty} \delta[n-1-k]$$

$x[n] = x_1[n] - x_2[n]$

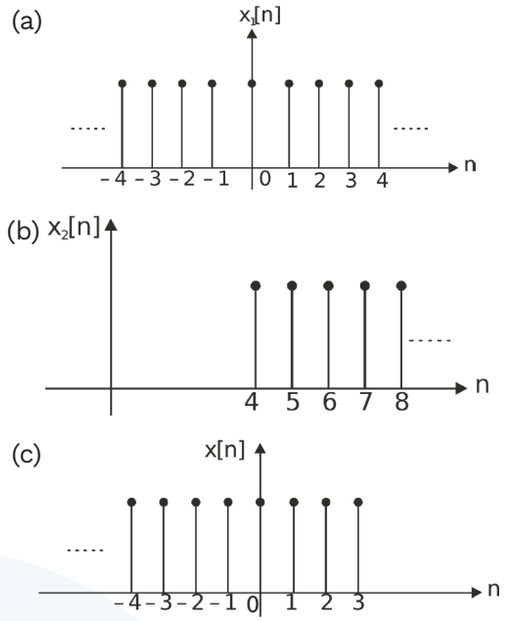


Fig. 1.21

x[n] = u[–n + 3]

x[n] = u[Mn – n_0]

M = –1 and n_0 = –3

Discrete Time System

It is a system for which both inputs and outputs are Discrete Time (DT) signals.

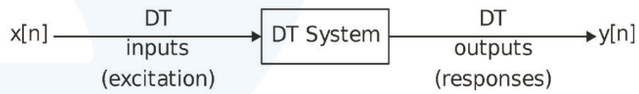
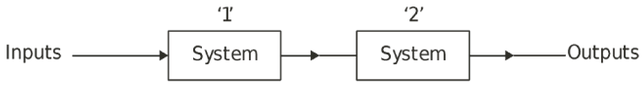


Fig. 1.22

Interconnection of Systems

A practical system may have many subsystems, which are interconnected in following ways.

1. Cascade (Series) Interconnection



(a)

Fig. 1.23

2. Parallel interconnection

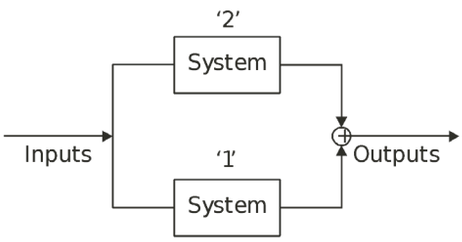


Fig. 1.24

3. Series-parallel interconnection

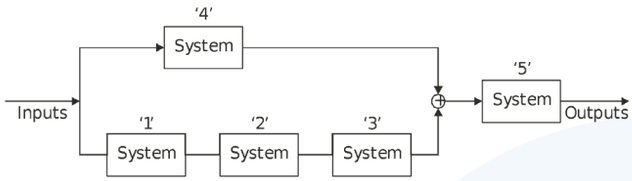


Fig. 1.25

4. Feedback interconnection

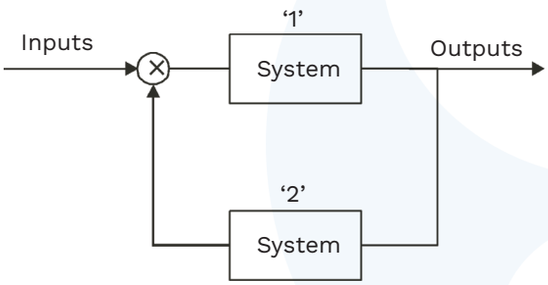


Fig. 1.26 System Interconnection

Basic Properties of System

Memory

A system for which value of the output signals depend on the past or future values of inputs signals, besides on the present value of the input signal, then the system is said to have memory, and is called dynamic system.

A system without memory means output signal, which does not depend on past or future values of input signals. It is called memoryless or static system.

System with Memory

$$\underset{\text{Present value of the out put signal}}{y[n]} = \underset{\text{Present value of the input signal}}{x[n]} + \underset{\text{Past value of the input signal}}{x[n-2]} + \underset{\text{Future value of the input signal}}{x[n+2]}$$

System without Memory (Memoryless)

$$\underset{\text{Present value of the output signal}}{y[n]} = \underset{\text{Present value of the input signal}}{x[n]} + 5$$

Stability

Bounded Input Bounded Output (BIBO) System

For every bounded input system, the output is bounded. In other words, the output of such a system does not diverge, if the input does not. Note that diverging of value means increasing or decreasing beyond any value. Mathematically, divergent $x[n] \to +\infty$ or $-\infty$ for some 'n'

For BIBO stability:

If $|x[n]| \le C < \infty$, for all n

then the system output

$|y[n]| \le D < \infty$, for all n

where C and D are some fixed positive constants.

Solved Examples

10. Identify if the following systems are BIBO stable or not.

a. $y[n] = (2n + 1)\, x[n]$

b. $y[n] = x[n] + x[n - 2] + x[n + 2]$

where x[n] is an input signal

Solution:

a. $\because |x[n]| \le c < \infty$

$|x[n]| \le C < \infty$, for all n for some fixed number C

$\therefore$ The input is bounded

$\therefore |y[n]| = (2n + 1)\, x|[n]|$

$|y[n]| = |2n + 1|\, |x[n]|$

$\Rightarrow |y[n]| \to +\infty$ as $n \to \pm\infty$

$\therefore$ Output y[n] diverges

Hence, the system is not BIBO stable, i.e., the system is unstable.

b. Given,

$y[n] = x[n] + x[n - 2] + x[n + 2]$

then input x[n] is bounded

$\Rightarrow |y[n]| = |x[n] + x[n - 2] + x[n + 2]|$

$\le |x[n]| + |x[n - 2]| + |x[n + 2]|$

$\leq C + C + C = 3C$

$|y[n]| \leq 3C < \infty$, for all n

$\Rightarrow$ The output y[n] remains bounded for all time, n

$\therefore$ The system is BIBO stable.

Causality

For a causal system, the present value of output depends on the present or past value of the input signal. On the other hand, for a non-causal system the output depends on one or more future values of input signal. For example:

- $y[n] = x[n-2] + x[n] \Rightarrow$ Causal system
- $y[n] = x[-n] \Rightarrow$ Non-causal system
- $y[n] = \frac{1}{2N+1}\sum_{k=-N}^{N} x[n-k] \Rightarrow$ Non-causal system

  $\Rightarrow$ A non-causal averaging system

Note:

- A causal system does not anticipate the future values of the input signal. So, it is often referred to as non-anticipative system.
- All memoryless systems are causal but not vice versa, for example:
  - y[n] = x[n], is memoryless, as well as, causal system
  - y[n] = x[n − 1] + x[n] is causal system but not memoryless

Linearity

1. **Superposition or additivity:** Consider a system initially at rest. Suppose, the system outputs are $y_1[n]$ and $y_2[n]$ corresponding to the input signals $x_1[n]$ and $x_2[n]$, respectively. Then, for the same system with input $x_1[n] + x_2[n]$ produces the output as $y_1[n] + y_2[n]$, the system is said to follow the principle of superposition.
2. **Homogeneity:** Consider a system initially at rest. Suppose, the system is excited with an input x[n] produces output as y[n]. Then, the system is said to follow the property of homogeneity, with an input ax(n) produces output as ay(n), i.e., the output y(n) is scaled by exactly the same factor 'a' as input. It also means that zero input yields zero output.

Mathematical Description

$$\boxed{ax_1[n] + bx_2[n] \longrightarrow ay_1[n] + by_2[n]}$$

Solved Examples

11. Consider a discrete time system described by the input-output relation.

$y[n] = n^2 x[n]$

Prove that the system is linear.

Solution:

Consider an input:

$$x[n] = \sum_{i=1}^{N} \alpha_i x_i[n]$$

Then, the resulting output signal of the system is:

$$y[n] = n^2 x[n]$$

$$= n^2 \sum_{i=1}^{N} \alpha_i x_i[n]$$

$$= \sum_{i=1}^{N} \alpha_i n^2 x_i[n]$$

$$y[n] = \sum_{i=1}^{N} \alpha_i y_i[n]$$

$\Rightarrow$ The system is linear.

Time Invariance

- For a time invariant system, the behaviour and characteristics of the system are fixed over time.
- If y[n] = H{x[n]}

  then y[n − k] = H{x(n − k)}
- Problem solving strategy: Compare the shifted output with the output produced by the shifted input

Linear Time Invariant (LTI) System

- If a system is linear, as well as, time invariant, then the system is called an LTI system.

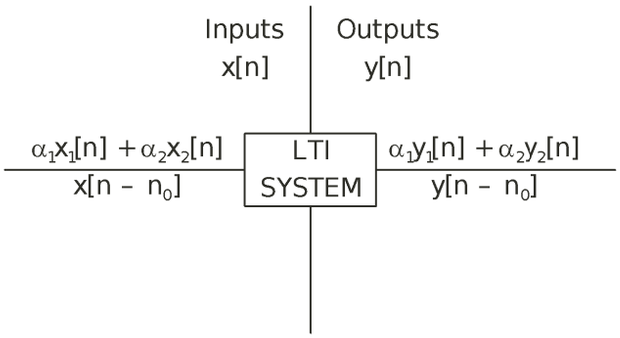
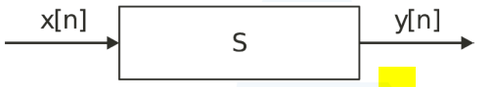


Fig. 1.27 LTI System

- A system, which does not satisfy either linearity, or time invariance property, or both is not an LTI system.

Invertible System

- A system is invertible if we determine its inputs from the output by any operation.
- Let for a system 'S'



If there exists any system S^{inv} such that:

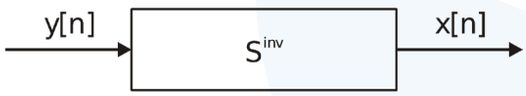


Fig. 1.28 Invertible System

In such a case, the system S is called an invertible system. If it is impossible to get back input signal x[n] from the output y[n] by any operation, then the system is called non-invertible system.
S^{Inv} is called inverse system of S.

Feedback System

- It is a special class of system where the output signal is feedback, which is added/ subtracted to the input signal. Look at the figure given below.

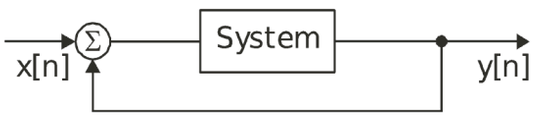


Fig. 1.29 Feedback System

Impulse Response of Linear Time Invariant System

The impulse response, h[n] of a discrete time LTI system (H) is defined as the response of the system when the input is $\delta[n]$.



i.e. $h[n] = H\{\delta(n)\}$

We follow a procedure similar to that used in the continuous time case by expressing an arbitrary input x[n] as the sum of impulse components. In discrete time case, the product of x[n] and a time shifted impulse sequence is $x[n]\delta[n-k] = x[k]\delta[n-k]$. The component of x[n] at n = k is $x[k]\delta[n-k]$. The multiplication of a signal by a time shifted impulse results in a time shifted impulse with amplitude given by the value of the signal at that time the impulse occurs.

The property allow us to express x[n] as the following weighted sum of time shifted impulses.

$x[n] = \ldots + x(-2)\ \delta(n+2) + x(-1)\ \delta(n+1) + x(0)\ \delta(n) + \ldots$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

- Decomposition of input signals into a linear combination of basis signals
  Weights $\Rightarrow$ Value of x[n] at time k
  Basis signals $\Rightarrow$ Time shifted impulses
- Let the operator H denote the system to which the input x[n] is applied. The output is
  $y[n] = H\{x[n]\}$

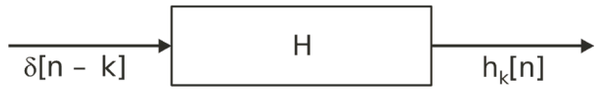
$$y[n] = H\left\{\sum_{k=-\infty}^{\infty} x[k]\delta[n-k]\right\}$$

- If the system is linear, then

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]H\{\delta[n-k]\}$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h_k[n]$$

Where,



- If the system is time invariant, i.e.,

$$h_k[n] = h[n-k] = H\{\delta[n-k]\}$$

$$\therefore\ y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$y[n] = x[n] * h[n]$ [convolution sum]

$\Rightarrow$ The output is weighted superposition of time shifted impulse responses

Procedures for finding convolution sum

Reflect and Shift Convolution Sum Evaluation

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y[n] = \sum_{k=-\infty}^{\infty} W_n[k]$$

where, $W_n[k] = x[k]\ h[n-k]$

Steps:

1. Draw the graph of $h[n-k]$ below the graph of $x[k]$
   Graph of $h[n-k]$
   $\because\ h[n-k] = h[-(k-n)]$
   Reflect $h[k]$ about $k = 0$ to get $h[-k]$, and then shift $h[-k]$ by $(-n)$. Note that if $n < 0$, then $h[-k]$ is shifted to the left and if $n > 0$, then $h[-k]$ is shifted to the right.
2. Begin with n large and negative.
3. Write the expression for $W_n[k]$
4. Increase n shifting $h[n-k]$ towards right till the expression of $W_n[k]$ changes.
   This change of $W_n[k]$ gives the beginning value of n for new set of $W_n[k]$
5. Repeat the steps 3 and 4 till all sets of n and the corresponding $W_n[k]$ are identified and written.
6. For each expression of $W_n[k]$, sum it from $k = -\infty$ to $k = +\infty$ and obtain $y[n]$

Solved Examples

12. For $x[n] = u[n]$ and $h[n] = \alpha^n u[n]$, $0 < \alpha < 1$ find the output $y[n] = x[n] \times h[n]$.

Solution:

$y[n] = x[n] \times h[n]$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Step 1:

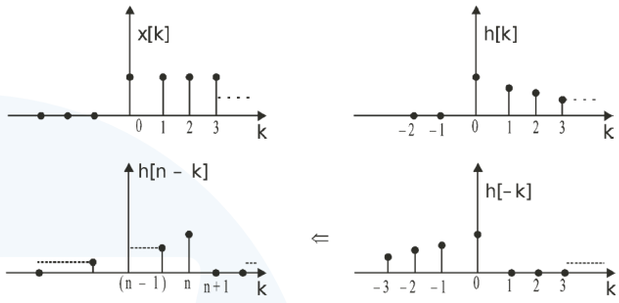


Fig. 1.29 (a)

Step 2:

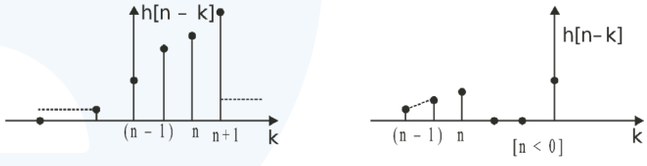


Fig. 1.29 (b)

Step 3–5:

1. $-\infty < n < 0$ $\quad W_n[k] = x[k]\ h[n-k]$
   $= 0$, for all k
2. $0 \le n < \infty$ $\quad W_n[k] = x[k]\ h[n-k]$
   $= u[k]\alpha^{(n-k)}\ u[n-k]$

$$W_n[k] = \begin{cases} \alpha^{(n-k)}, & 0 \le k \le n \\ 0, & \text{otherwise} \end{cases}$$

Step 6: Finding the output $y[n]$

1. $-\infty < n < 0$

$$y[n] = \sum_{k=-\infty}^{\infty} W_n[k] = 0$$

2. $0 \le n < \infty$

$$y[n] = \sum_{k=-\infty}^{\infty} W_n[k]$$

$$= \sum_{k=0}^{n} \alpha^{(n-k)}$$

$$= \alpha^n \sum_{k=0}^{n} \alpha^{-k}$$

$$= \alpha^n (1 + \alpha^{-1} + \alpha^{-2} + \ldots + \alpha^{-n})$$

$$= \alpha^n \left(\frac{1 - (\alpha^{-1})^{(n+1)}}{1 - \alpha^{-1}} \right)$$

$$y[n] = \frac{\alpha^{(n+1)} - 1}{(\alpha - 1)} = \frac{1 - \alpha^{(n+1)}}{(1 - \alpha)}$$

Tabular Method for Finite Length Discrete Time Signals x[n] and h[n]

- In this method, the convoluted sequence will have in total (n + m – 1) terms, where n and m are the number of terms in the sequence to be convolved.

Solved Examples

13. For the input signal,

$x[n] = \delta[n+1] - 2\delta[n] + 3\delta[n-1]$

and impulse response,

$h[n] = \delta[n] + 4\delta[n-1] - 2\delta[n-2] + \delta[n-3]$

Calculate the output, y[n] = x[n] × h[n]

Solution:

Note that, $x[n] = \left\{1, \underset{\uparrow}{-2}, 3\right\}$

$h[n] = \left\{\underset{\uparrow}{1}, 4, -2, 1\right\}$

Step 1: Form the table

x[n] / h[n]	1	4	–2	1
1	1	4	–2	1
–2	–2	–8	4	–2
3	3	12	–6	3

↑
Values of x[n]

Table 1.1

The underlined elements correspond to n = 0, before it [elements corresponding to (n < 0)] and after it [element corresponding to (n > 0)] elements.

Step 2:

y[0] = x[–1] h[1] + x[0] h[0]
= 4 + (–2) = Sum of diagonal elements
= 2

Similarly, other elements:

$y[n] = \{1, 4+(-2), (-2)+(-8)+3, 1+4+12, (-2)+(-6), 3\}$

$y[n] = \left\{1, \underset{\uparrow}{2}, -7, 17, -8, 3\right\}$

Note: This can also be evaluated by applying the convolution sum directly.

This method is suitable for response evolution only when both the signals have finite length.

Properties of Convolution Sum

1. Commutative property:
   $x(n) \times h(n) = h(n) \times x(n)$
2. Associative property:
   $x(n)[h_1(n) \times h_2(n)] = [x(n) \times h_1(n)] \times h_2(n)$
3. Distributive property:
   $x(n) \times [h_1(n) + h_2(n)] = x(n) \times h_1(n) + x(n) \times h_2(n)$
4. Shifting property:
   $x(n) \times h(n) = y(n)$, then
   $x(n) \times h(n - n_0) = x(n - n_0) \times h(n) = y(n - n_0)$
   and $x(n - n_1) \times h(n - n_2) = y(n - n_1 - n_2)$
5. Convolution with impulse: Convolution of a signal x(n) with a unit impulse function results signal x(n) itself.
   $x(n) * \delta(n) = x(n)$
6. **Width property:** The number of samples in the non-zero extent of a discrete convolution of two sequences is one lesser than the sum of the number of samples in the non-zero extent of the two sequences. If N_x & N_h are the number of samples in the non-zero extent of the sequences x(n) & h(n) respectively, then the number of samples in the non-zero extent of the sequence y(n) = x(n) * h(n) is:

   $N_y = N_x + N_h - 1$

7. The sum of the impulse in a convolution sum of two discrete time sequence is the product of the sums of the impulses in the two individual sequences. Let the sum of all the impulse in the signals y(n), x(n) & h(n) be S_y, S_x & S_h respectively. Then

Proof:

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

$$S_y = \sum_{n=-\infty}^{\infty} y(n) = \sum_{n=-\infty}^{\infty}\sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

$$S_y = \sum_{k=-\infty}^{\infty} x(k) \sum_{n=-\infty}^{\infty} h(n-k)$$

(Interchanging the order)

$$S_y = S_x S_h$$

Interconnection of LTI Systems

Parallel Connections

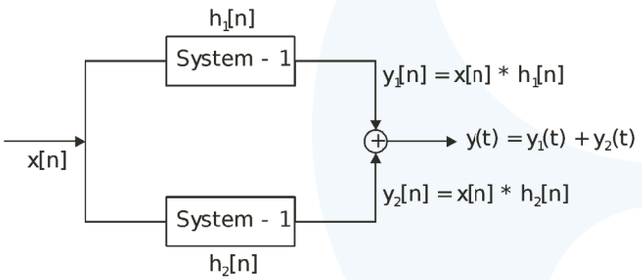


Fig. 1.30

$$y[n] = x[n] \times h_1[n] + x[n] \times h_2[n]$$

$$= x[n] \times \{h_1[n] + h_2[n]\}$$

$$y[n] = x[n] \times h[n]$$

Where,

$$h[n] = h_1[n] + h_2[n]$$

= Impulse response of the equivalent system

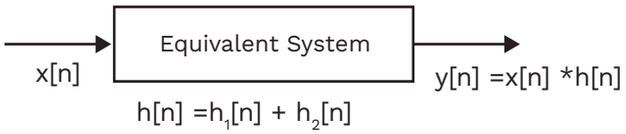


Fig. 1.31

Series or Cascade Connections

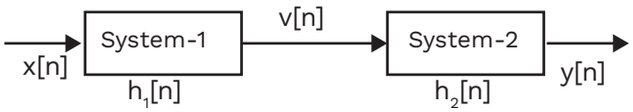


Fig. 1.32

Where,

$$v[n] = x[n] \times h_1[n]$$

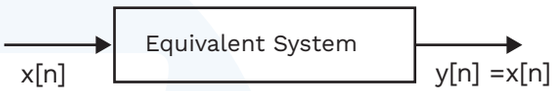
$$y[n] = v[n] \times h_2[n] \text{ and}$$

$$y[n] = \{x[n] \times h_1[n]\} \times h_2[n]$$

$$= x[n] \times \{h_1[n] \times h_2[n]\}, \text{ (Prove it)}$$

$$= x[n] \times \{h[n]\} \quad [\because h[n] = h_1[n] \times h_2[n]]$$

= Impulse response of the equivalent system



$$h[n] = h_1[n] * h_2[n]$$

$$= h_2[n] * h_1[n] \text{ (commutative property)}$$

Step Response of LTI System

The step response of (Discrete Time) LTI system is defined to be the response of the system when the input is u[n].

$$s[n] = H\{u[n]\}$$

$$= h[n] \times u[n]$$

$$= \sum_{k=-\infty}^{\infty} h[k]u[n-k]$$

$$s[n] = \sum_{k=-\infty}^{n} h[k]$$

$$\Rightarrow h[n] = s[n] - s[n-1]$$

Impulse Response and LTI System Characteristics

With Memory or Memoryless

For discrete time memoryless LTI system: $h[n] = k\delta[n]$

With memory LTI system: $h[n_0] \neq 0$ for $n_0 \neq 0$

Causality

By definition, a causal system does not respond to an input event until that event actually occurs. Therefore, we must have:

$h[n] = 0,\ n < 0$

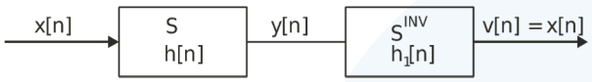
Stability

- Condition for BIBO stable LTI systems
- For discrete time LTI stable system yields.

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

i.e., the impulse response is absolutely summable

Invertibility and Deconvolution



Output of the system,

$y[n] = x[n] \times h[n]$

Output of the inverse system, Or

$v[n] = [x[n] \times h[n]] \times h_1[n]$

$v[n] = x[n] \times [h[n] \times h_1[n]]$

It's condition for an invertible system

$h[n] \times h_1[n] = \delta[n]$

$v[n] = x[n] \times \delta[n] = x[n]$

Deconvolution

Observe the operation performed by the inverse system. It recovers (x[n]) from (x[n] × h[n]), this process is called deconvolution.

Solved Examples

14. Show that if the input x[n] to a discrete time LTI system is periodic with period N, then the output y[n] is also periodic with period N.

Solution:

For a discrete time LTI system with impulse response, the output due to the input x[n] is:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k];\ \text{for all } n$$

Replace n by (n + N), we get

$$y[n+N] = \sum_{k=-\infty}^{\infty} h[k]x[(n+N)-k] = \sum_{k=-\infty}^{\infty} h[k]x[(n-k)+N]$$

$\because$ x[n] is periodic with period N

$\therefore$ $x[(n-k)+N] = x[n-k]$

$$\therefore\ y[n+N] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = y[n]$$

$\Rightarrow$ y[n] is periodic with the same period N

15. Evaluate the step response for the LTI system represented by the following impulse responses.

a. $h(n) = \left(-\frac{1}{2}\right)^n u(n)$

b. $h(n) = \delta(n) - \delta(n-5)$

Solution:

a. $h(n) = \left(-\frac{1}{2}\right)^n u(n)$

Step Response:

$$s(n) = \sum_{k=-\infty}^{n} h(k)$$

$$s(n) = \sum_{k=-\infty}^{n} \left(\frac{-1}{2}\right)^k u(k)$$

$$= \sum_{k=0}^{n} \left(\frac{-1}{2}\right)^{-k}$$

$$= \frac{1-\left(\frac{-1}{2}\right)^{n+1}}{1-\left(\frac{-1}{2}\right)} = \frac{2}{3}\left[1-\left(\frac{-1}{2}\right)^{n+1}\right]$$

$$s(n) = \frac{2}{3}\left[1-\left(\frac{-1}{2}\right)^{n+1}\right]$$

b. $h(n) = \delta(n) - \delta(n-5)$

Step Response:

$$s(n) = \sum_{k=-\infty}^{n} h(k)$$

$$= \sum_{k=-\infty}^{n} \delta(k) - \sum_{k=-\infty}^{n} \delta(k-5)$$

$$s(n) = u(n) - u(n-5)$$

16. Determine the overall impulse response of the system shown below.

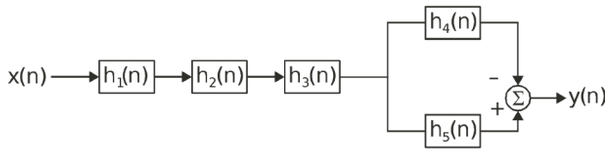


Fig. 1.33

Given that,

$h_1(n) = \delta(n) - a\,\delta(n-1)$

$h_2(n) = \left(\frac{1}{2}\right)^n u(n)$

$h_3(n) = a^n u(n)$

$h_4(n) = (n-1)\,u(n)$

$h_5(n) = \delta(n) + n\,u(n-1) + \delta(n-2)$

Solution:

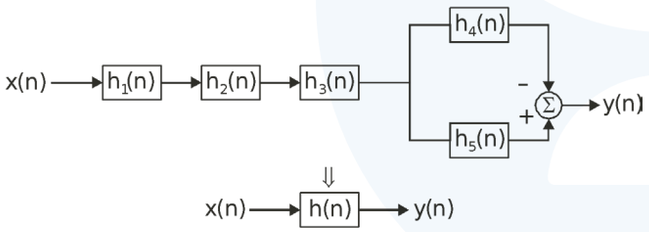


Fig. 1.34

$$h(n) = [h_1(n) \times h_2(n)] \times h_3(n) \times \{h_5(n) - h_4(n)\}$$

$$h(n) = [h_1(n) \times h_3(n)] \times h_2(n) \times \{h_5(n) - h_4(n)\}$$

$$= [\{\delta(n) - a\delta(n-1)\} \times a^n u(n)] \times h_2(n)$$

$$\times \{\delta(n) + nu(n-1) + \delta(n-2) - (n-1)u(n)\}$$

$$= \left[\{\delta(n) - a\,\delta(n-1)\} \times a^n u(n)\right] \times h_2(n)$$

$$= \left[\{\delta(n) \times a^n u(n)\} - \{a\delta(n-1) \times a^n u(n)\}\right]$$

$$\times h_2(n) \times \{\delta(n) + \delta(n-2) + u(n)\}$$

$$= \left[a^n u(n) - a^n u(n-1)\right] \times h_2(n)$$

$$\times \{\delta(n) + \delta(n-2) + u(n)\}$$

$$= a^n[u(n) - u(n-1)] \times [\{h_2(n) \times \delta(n)\} + h_2(n)$$

$$\times \delta(n-2) + h_2(n) \times u(n)]$$

$$= \delta(n) \times h_2(n) + \delta(n) \times h_2(n-2) + \delta(n)$$

$$\times h_2(n) \times u(n) \{\because a^n \delta(n) = \delta(n)\}$$

$$= h_2(n) + h_2(n-2) + \sum_{k=-\infty}^{\infty} h_2(k)u(n-k)$$

$$= \left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{2}\right)^{n-2} u(n-2) + \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^k u(k)u(n-k)$$

$$= \left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{2}\right)^{n-2} u(n-2) + \sum_{k=0}^{n} \left(\frac{1}{2}\right)^k$$

$$\sum_{k=0}^{n} \left(\frac{1}{2}\right)^k = \left[\frac{1-\left(\frac{1}{2}\right)^{n+1}}{1-\left(\frac{1}{2}\right)}\right], n \geq 0$$

$$= 2\left[1-\left(\frac{1}{2}\right)^{n+1}\right], n \geq 0$$

$$= 2u(n) - 2\left(\frac{1}{2}\right)^{n+1} u(n)$$

$$= 2u(n) - \left(\frac{1}{2}\right)^n u(n)$$

$$h(n) = \left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{2}\right)^{n-2} u(n-2) + \sum_{k=0}^{n} \left(\frac{1}{2}\right)^k$$

$$= \left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{2}\right)^{n-2} u(n-2) + 2u(n) - 2\left(\frac{1}{2}\right)^{n+1} u(n)$$

$$h(n) = \left(\frac{1}{2}\right)^{n-2} u(n-2) + 2u(n)$$

Practice Exercise

Level I

1. The signal u[n] + u[−n] is equal to
(A) 2
(B) $1+\delta[n]$
(C) $2+\delta[n]$
(D) 1

2. The even part of a function x[n] = u[n] − u[n − 4] is:
(A) $\frac{1}{2}\{1+\delta[n]-u[n-4]-u[-n-4]\}$
(B) $\frac{1}{2}\{u[n+3]-u[n-4]+\delta[n]\}$
(C) $\frac{1}{2}\{u[n]+u[-n]-u[n-4]-u[-n-4]\}$
(D) All of the above

3. The energy of a signal $A\delta[n]$ is:
(A) A^2
(B) $\frac{A^2}{2}$
(C) $\frac{A^2}{4}$
(D) 0

4. The energy of a signal nu[n] is:
(A) $\frac{n(n+1)}{2}$
(B) $\frac{n(n+1)(2n+1)}{6}$
(C) $\left(\frac{n(n+1)}{2}\right)^2$
(D) ∞

5. The power of a signal u[n] is:
(A) n
(B) 1
(C) 1/2
(D) ∞

6. Signals that are generated by multiple sources are called:
(A) multichannel signals
(B) multidimensional signals
(C) one-dimensional signal
(D) one-channel signal

7. A signal that is the function of a single independent variable is called:
(A) multichannel signal
(B) multidimensional signal
(C) one-dimensional signal
(D) one-channel signal

8. A signal that is the function of two or more independent variables is called:
(A) multichannel signal
(B) multi-dimensional signal
(C) one-dimensional signal
(D) one-channel signal

9. Let u(n) be a unit step signal. The signal $u(N_0-n)$ can be described as:
(A) $x(n)=\begin{cases}1 & n<N_0\\ 0 & \text{otherwise}\end{cases}$
(B) $x(n)=\begin{cases}1 & n\le N_0\\ 0 & \text{otherwise}\end{cases}$
(C) $x(n)=\begin{cases}1 & n>N_0\\ 0 & \text{otherwise}\end{cases}$
(D) $x(n)=\begin{cases}1 & n\ge N_0\\ 0 & \text{otherwise}\end{cases}$

10. Given a unit step signal u(n), the time difference u(n) −u(n − 1) is equal to:
(A) a unit impulse signal
(B) another step signal
(C) a unit ramp signal
(D) None of the above

11. If a periodic signal x(n) of period N_1 is added to another periodic signal of period N_2, then the period of the resulting signal is always:
(A) $N_1 + N_2$
(B) $N_1 N_2$
(C) LCM of N_1 and N_2
(D) GCD of N_1 and N_2

12. The system having input x(n) related to output y(n) as $y(n) = \log_{10}|x(n)|$ is:

(A) non-linear, causal, and stable
(B) linear, non-causal, and stable
(C) non-linear, causal, and not stable
(D) linear, non-causal, and not stable

13. To obtain x(4 – 2n) from the given signal x(n), which of the following precedence (or priority) rule is used for operations on the independent variable n.
(A) Time scaling → Time shifting → Reflection
(B) Reflection → Time scaling → Time shifting
(C) Time scaling → Reflection → Time shifting
(D) Time shifting → Time scaling → Reflection

14. The system described by y(n) = nx(n) is:
(A) linear, time varying, and stable
(B) non-linear, time invariant, and unstable
(C) non-linear, time varying, and stable
(D) linear, time varying, and unstable

15. The period of the signal $x(n) = \cos\left(\frac{\pi}{2}n\right) \cos\left(\frac{\pi}{4}n\right)$ is:
(A) 8
(B) 32
(C) 4
(D) 12

16. The signal $x(n) = \sum_{n=-\infty}^{\infty} \delta(n)$ is:
(A) ∞
(B) 1
(C) 0
(D) Undefined

17. $x(n) = a^{|n|}$, $|a| < 1$ is:
(A) an energy signal
(B) a power signal
(C) neither energy nor power signal
(D) energy, as well as, power signal

18. The system characterised by the equation y(n) = x(n) + b is:
(A) linear for b = 0
(B) linear if b > 0
(C) linear if b < 0
(D) non-linear

19. The discrete time signal $x(n) = (-1)^n$ is periodic with the fundamental period _____.
(A) 6
(B) 2
(C) 4
(D) 0

20. A useful property of the unit impulse $\delta(n)$ is that:
(A) $\delta(kn) = k\delta(n)$
(B) $\delta(kn) = \delta(n)$
(C) $\delta(kn) = \frac{1}{|k|}\delta(n)$
(D) none of these

21. Any signal x(n) with even part $x_e(n)$ and odd part $x_0(n)$, can be represented as:
(A) $x_e(n) + x_0(n)$
(B) $x_e(n) - x_0(n)$
(C) $\frac{x_e(n)}{x_0(n)}$
(D) $x_e(n) \times x_0(n)$

22. Periodic signals are:
(A) x(n + N) = x(n)
(B) x(n – N) = x(n)
(C) x(n + mN) = x(n), m integer
(D) All of the above

23. Invertible system are those systems, where:
(A) the output can be uniquely obtained from the knowledge of input
(B) the input signal can be uniquely determined by observing the output signal
(C) Both (A) and (B)
(D) the system output is always constant

24. Choose the false statement.
(A) $n\delta(n) = 0$
(B) $u(n) = \sum_{k=-\infty}^{n} \delta(k)$
(C) $\delta(n) = \sum_{k=-\infty}^{n} u(k)$
(D) $n^2\delta(n-2) = 4\delta(n-2)$

25. The system described by y(n) = cos[x(n)] is:
(A) linear and invertible
(B) linear and non-invertible
(C) non-linear and invertible
(D) non-linear and non-invertible

26. Three signals $x_1(t) = \cos(6\pi t)$, $x_2(t) = \cos(14\pi t)$, and $x_3(t) = \cos(26\pi t)$ are sampled at the rate of 10 Hz. Let the resulting signal be $x_1(n)$, $x_2(n)$, and $x_3(n)$. Then:
(A) $x_1(n) \neq x_2(n) \neq x_3(n)$
(B) $x_1(n) = x_2(n) = x_3(n)$
(C) $x_1(n) = x_2(n)$, but $x_3(n)$ is different
(D) $x_2(n) = x_3(n)$, but $x_1(n)$ is different

Level II

1. The signal $x[n] = \cos\frac{\pi n}{9} + \sin\left(\frac{\pi n}{7} + \frac{1}{2}\right)$ is:
(A) periodic with period N = 126
(B) periodic with period N = 32
(C) periodic with period N = 252
(D) not periodic

2. The signal $x[n] = \cos\left(\frac{n}{8}\right)\cos\left(\frac{\pi n}{8}\right)$ is:
(A) periodic with period 16π
(B) periodic with period $16(\pi + 1)$
(C) periodic with period 8
(D) not periodic

3. The signal
$x[n] = \cos\left(\frac{\pi n}{2}\right) - \sin\left(\frac{\pi n}{8}\right) + 3\cos\left(\frac{\pi n}{4} + \frac{\pi}{3}\right)$ is:
(A) periodic with period 16
(B) periodic with period 4
(C) periodic with period 2
(D) not periodic

4. The signal $x[n] = 2e^{j\left(\frac{n}{6} - \pi\right)}$ is:
(A) periodic with 12π
(B) periodic with 12
(C) periodic with 11π
(D) not periodic

5. The sinusoidal signal has fundamental period N = 10 samples. The smallest angular frequency, for which x[n] is periodic, is:
(A) $\frac{1}{10}$ rad / cycle
(B) 10 rad/cycle
(C) 5 rad/cycle
(D) $\frac{\pi}{5}$ rad / cycle

6. Let $x[n], -5 \le n \le 3$ and $h[n], 2 \le n \le 6$ be two finite duration signal. The range of their convolution is:
(A) $-7 \le n \le 9$
(B) $-3 \le n \le 9$
(C) $2 \le n \le 3$
(D) $-5 \le n \le 6$

7. Consider the sequence:
x[n] = [−4 − j5 1 + j2 4]
The conjugate anti-symmetric part of the sequence is:
(A) [−4 −j2.5 j24 − j2.5]
(B) [−j2.51 j2.5]
(C) [−j5 j2 0]
(D) [−4 1 4]

8. Let P be linearity, Q be time invariance, R be causality, and S be stability. A discrete time system has the input output relationship.

$$y(n) = \begin{cases} x(n), & n \ge 1 \\ 0, & n = 0 \\ x(n+1), & n \le -1 \end{cases}$$

Where, x(n) is the input and y(n) is the output. The above system has the properties:
(A) P and S, but not Q and R
(B) P, Q, and S, but not R
(C) P, Q, R, and S
(D) Q, R, and S, but not P

9. The output y(n) of a discrete time system for an input x[n] is $y[n] = \max_{-\infty \le k \le n} |x[k]|$
The unit impulse response of the system is:
(A) unit step signal u[n]
(B) 0 for all n
(C) unit impulse signal $\delta[n]$
(D) 1 for all n

10. Consider a single input single output discrete time system with x[n] as input and y[n] as output where the two are related as:

$$y(n) = \begin{cases} n|x[n]| & \text{for } 0 \le n \le 10 \\ x[n] - x[n-1] & \text{otherwise} \end{cases}$$

Which one of the following statements is true about the system?

(A) It is causal and stable.
(B) It is causal but not stable.
(C) It is not causal but stable.
(D) It is neither causal nor stable.

11. A discrete time signal $x[n] = \sin\left(\pi^2 n\right)$, n being an integer is:

(A) periodic with period π
(B) periodic with period π^2
(C) periodic with period $\pi / 2$
(D) not periodic

12. A system is defined by its impulse response $h(n) = 2^n u(n - 2)$. The system is:

(A) stable and causal
(B) causal but not stable
(C) stable but not causal
(D) unstable and not causal

13. Two discrete time systems with impulse responses $h_1[n] = \delta[n-1]$ and $h_2[n] = \delta[n-2]$ are connected in cascade. The overall impulse response of the cascaded system is:

(A) $\delta[n-1] + \delta[n-2]$
(B) $\delta[n-4]$
(C) $\delta[n-3]$
(D) $\delta[n-1]\,\delta[n-2]$

14. The impulse response h[n] of a linear time invariant system is given by:
h(n) = u[n + 3] + u[n – 2] – 2u[n – 7]
Where, u[n] is the unit step sequence. The above system is:

(A) stable but not causal
(B) stable and causal
(C) causal but unstable
(D) unstable and not causal

15. Two discrete time signals x[n] and h[n] are both non-zero only for n = 0, 1, 2 and are zero otherwise. It is given that:
x[0] = 1, x[1] = 2, x[2] = 1, h(0) = 1
Let y[n] be the linear convolution of x[n] and h[n]. Given that y[1] = 3 and y[2] = 4, the value of the expression (10y[3] + y[4]) is _____.

16. Consider a discrete time signal.

$$x[n] = \begin{cases} n & \text{for } 0 \le n \le 10 \\ 0 & \text{otherwise} \end{cases}$$

If y[n] is the convolution of x[n] with itself, the value of y[4] is _____.

17. Let y[n] denote the convolution of h[n] and g[n], where $h[n] = (1/2)^n u(n)$ and g[n] is a causal sequence. If y[0] = 1 and y[1] = 1/2, then g[1] equals

(A) 0
(B) 1/2
(C) 1
(D) 3/2

18. A discrete time linear shift invariant system has an impulse response h[n] with h[0] = 1, h[1] = –1, h[2] = 2 and zero otherwise. The system is given an input sequence x[n] with x[0] = x[2] = 1 and zero otherwise. The number of non-zero samples in the output sequence y[n] and the value of y[2] are, respectively:

(A) 5, 2
(B) 6, 2
(C) 6, 1
(D) 5, 3

19. A system with input x[n] and output is given as:

$$y[n] = \left(\sin\frac{5}{6}\pi n\right) x(n)$$

The system is:

(A) linear, stable, and invertible
(B) non-linear, stable, and non-invertible
(C) linear, stable, and non-invertible
(D) linear, unstable, and invertible

20. Given h[n] = [1, 2, 2], f[n] is obtained by convolution h[n] with itself and g[n] by correlating h[n] with itself. Which one of the following statements is true?

(A) f[n] is causal and its maximum value is 9
(B) f[n] is non-causal and its maximum value is 8

(C) g[n] is causal and its maximum value is 9

(D) g[n] is non-causal and its maximum value is 9

21. Consider the discrete time signal:

$x(n) = \left(\frac{1}{3}\right)^n u(n)$, where $u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$.

Define the signal $y(n)$ as $y(n) = x(-n), -\infty < n < \infty$.

Then $\sum_{n=-\infty}^{\infty} y(n)$ equals.

(A) $-\frac{2}{3}$

(B) $\frac{2}{3}$

(C) $\frac{3}{2}$

(D) 3

Answer Key

Level I

1. (B)	**2.** (D), (A)	**3.** (A)	**4.** (D)	**5.** (C)	**6.** (A)	**7.** (C)	**8.** (B)	**9.** (B)	**10.** (A)
11. (C)	**12.** (A)	**13.** (D)	**14.** (D)	**15.** (A)	**16.** (C)	**17.** (A)	**18.** (A)	**19.** (C)	**20.** (B)
21. (A)	**22.** (D)	**23.** (B)	**24.** (C)	**25.** (D)	**26.** (B)				

Level II

1. (A)	**2.** (D)	**3.** (A)	**4.** (D)	**5.** (D)	**6.** (B)	**7.** (A)	**8.** (A)	**9.** (A)	**10.** (A)
11. (D)	**12.** (A)	**13.** (C)	**14.** (A)	**15.** 31	**16.** 10	**17.** (A)	**18.** (D)	**19.** (C)	**20.** (D)